

11.2 Fundamental Particles

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.2 Fundamental Particles
Difficulty	Medium

Time allowed: 30

Score: /25

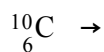
Percentage: /100

Question 1a

Carbon-10 ($^{10}_6\text{C}$) is an isotope that decays to an isotope of boron (symbol **B**) by the emission of a β^+ particle.

(i)

Complete the nuclear equation for this decay, including the proton and nucleon numbers of all the particles involved.



[3]

(ii)

During the decay, a quark in the carbon-10 nucleus changes into a different type.

State the change of quark that occurs.

[1]

[4 marks]

Question 1b

State the two leptons in this decay.

[2 marks]

Question 1c

State and explain why a hadron with charge $-3e$ cannot exist.

e is the elementary charge.

[3 marks]

Question 1d

State the possible quark combination for a hadron of charge $+2e$.

[1 mark]

Question 2a

State the quark composition of an alpha particle.

[2 marks]

Question 2b

Use the quark model to show that the charge on the anti-neutron is 0.

[2 marks]

Question 2c

A sigma baryon Σ is a hadron that consists of an up (u) quark, a down (d) quark and a strange (s) quark.

State the quark composition of Σ^+ , Σ^0 and Σ^- .

[3 marks]

Question 2d

Explain why a baryon with the quark composition $u\bar{d}s$ cannot exist.

[1 mark]

Question 3a

State the number of 'up' quarks in the nucleus of ${}_{13}^{27}\text{Al}$.

[1 mark]

Question 3b

The **D** particle has a quark structure $u\bar{c}$.

Determine

(i)
the classification of the **D** particle,

[1]

(ii)
the charge on the **D** particle.

[1]

[2 marks]

Question 3c

Using the quark model, show that the magnitude of the charge of a proton is equal to the charge of an electron.

[1 mark]

Question 3d

Using the quark model for β^+ decay, show that charge is conserved in this decay.

[3 marks]

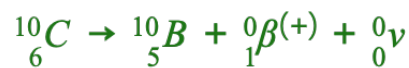


Model Answers: Medium

1a

(a)

(i) The nuclear equation for the decay is:



- ${}^1_5\text{B}$ [1 mark]
- ${}^0_1\beta^{(+)} / {}^0_1e^{+}$ [1 mark]
- ${}^0_0\nu$ [1 mark]

(ii) The change of quark is:

- Up quark to down quark; [1 mark]

[Total: 4 marks]

Although you have not been given the proton and nucleon numbers of the boron nucleus, you can figure these out as the top row of numbers and the bottom row of numbers must **balance** on both sides of the equation.

For example, on the left-hand side, the proton number is 6. On the right-hand side, the proton number is $1 + 0 = 1$. Therefore, the proton number of boron must be 5 so both sides have a total of 6.

1b

(b) The two leptons in the decay are:

- Beta-plus particle / β^{+} / positron; [1 mark]
- (Electron) neutrino; [1 mark]

[Total: 2 marks]

Make sure to say it is a **positron** and not an electron in this decay!

1c

(c) A hadron with a charge of $-3e$ cannot exist because:

Any **three** from:

- Quarks can either have a charge of $\pm\frac{1}{3}e$ or $\pm\frac{2}{3}e$; [1 mark]
- A meson contains a quark and an anti-quark; [1 mark]
- (Hence) mesons can only have a charge of 0 or $\pm 1e$ (therefore it cannot be a meson); [1 mark]
- A baryon contains three (anti)quarks; [1 mark]
- (Hence) baryons can only have a charge of 0 or $\pm 2e$ (therefore, it cannot be a baryon); [1 mark]

[Total: 3 marks]

Remember, a hadron is any particle made up of quarks, so make sure you refer to both mesons and baryons

1d

(d) A possible quark combination for a hadron of charge $+2e$ is:

Any **one** from:

- uuu / three up quarks [1 mark]
- ttt / three top quarks [1 mark]
- ccc / three charm quarks [1 mark]
- utc / up quark, top quark, charm quark [1 mark]
- $\bar{d}\bar{s}\bar{b}$ / anti-down quark, anti-strange quark, anti-bottom quark [1 mark]

[Total: 1 mark]

Although a meson is a type of hadron, a meson cannot have a charge of $+2e$ as there is no combination of a quark and anti-quark pair that gives a charge of $+2e$.

It is important you memorise the **charges** of each of the quarks, and remember that the equivalent anti-quark has the **opposite** charge.

2a

(a) The quark composition of an alpha particle is:

- (An alpha particle has) 2 protons (uud) and 2 neutrons (udd); [1 mark]
- (So its quark composition is) 6 up (u) **AND** 6 down (d) quarks; [1 mark]

[Total: 2 marks]

A proton would have a quark composition of uud . A neutron has udd . Therefore, $uud + uud + udd + udd + udd + udd = 6$ up and 6 down quarks.

2b

(b) Use the quark model to show that the charge on the anti-neutron is 0

- (The quark structure) is anti-up anti-down anti-down / $\bar{u}\bar{d}\bar{d}$; [1 mark]
- $-\frac{2}{3}(e) + \frac{1}{3}(e) + \frac{1}{3}(e) = 0$ [1 mark]

[Total: 2 marks]

It is important you remember all the charges of each quark. An anti-quark is going to have the same magnitude but **opposite charge**.

2c

(c) State the quark composition of Σ^+ , Σ^0 and Σ^- :

- $\Sigma^+ = uus$; [1 mark]
- $\Sigma^0 = uds$; [1 mark]
- $\Sigma^- = dds$; [1 mark]

[Total: 3 marks]

The subscript for each of the Σ particles indicates its charge. Since it is a **baryon**, this means it is made up of **three quarks** (anti-baryons are made up of three antiquarks). It is important to remember the charge of each of these quarks to combine them in a way to get the total charge required:

- $uus = +\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1$
- $uds = +\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$
- $dds = -\frac{1}{3} - \frac{1}{3} - \frac{1}{3} = -1$

2d

(d) A baryon with the quark composition $u\bar{d}s$ cannot exist because:

- A baryon can only be made up of three quarks or three anti-quarks **OR** a baryon cannot consist of a mix of quarks and anti-quarks; [1 mark]

[Total: 1 mark]

3a

(a) The number of 'up' quarks in the nucleus of ${}_{13}^{27}\text{Al}$ is:

Determine the number of protons and neutrons:

- There are 13 protons
- There are $27 - 13 = 14$ neutrons

Calculate the number of up quarks:

- A proton (uud) has 2 up quarks and a neutron (udd) has 1 up quark
- Number of up quarks = $(13 \times 2) + (14 \times 1) = 40$ [1 mark]

[Total: 1 mark]

3b

(b)

(i) The classification of the D particle is:

- A meson; [1 mark]

(b)

(i) The classification of the D particle is:

- A meson; [1 mark]

(ii) The charge on the D particle is:

- $u\bar{c} = \frac{2}{3} + \left(-\frac{2}{3}\right) = 0$ [1 mark]

[Total: 2 marks]

Just 'hadron' will not be enough to get the mark for part (i), you should be as specific as possible.

3c

(c) Using the quark model, show that the magnitude of the charge of a proton is equal to the charge of an electron

- uud (2 up quarks and 1 down quark)
- $+\frac{2}{3}e + \frac{2}{3}e + \left(-\frac{1}{3}e\right) = 1e$ [1 mark]

[Total: 1 mark]

3d

(d) Using the quark model for β^+ decay, prove that charge is conserved

- β^+ decay is: $p \rightarrow n + e^+ + \nu_e$
- Therefore, the quark changes are: $u \rightarrow d + e^+ + \nu_e$ [1 mark]

Compare charges on both sides:

- $+\frac{2}{3} = -\frac{1}{3} + (+1) + 0 = +\frac{2}{3}$ [1 mark]
- (LHS = RHS), therefore, the charge is conserved; [1 mark]